

Gabriel Dorfsman-Hopkins

Assistant Professor of Mathematics | Co-Founder and Co-Director of the Explore Lab Makerspace
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EMPLOYMENT

St. Lawrence University Assistant Professor of Mathematics	2022-
University of California, Berkeley NSF RTG Postdoctoral Scholar	2019-2022
Institute for Computational and Experimental Research in Mathematics Postdoctoral Scholar	2019

EDUCATION

PhD, Mathematics, University of Washington Advisor: Max Lieblich Dissertation: Projective Geometry for Perfectoid Spaces	2019
BA, Mathematics, Dartmouth College	2013

RESEARCH AND SCHOLARSHIP

My scholarship spans algebraic geometry, arithmetic geometry, mathematical visualization, and interdisciplinary mathematical art. A central theme of my work is the use of visualization, fabrication, and artistic practice as tools for research, pedagogy, and mathematical communication. Much of this work is collaborative with undergraduate students, indicated below italics.

Mathematical Illustration and Visualization

On the Importance of Illustration for Mathematical Research Notices Amer. Math. Soc., 71.01, 105-115 With R. Coulon, E. Harriss, M. Skrodzki, K.E. Stange, G. Whitney. Cover Article	2024
Deformation Spaces and Static Animations Proc. Symp. Appl. Math. 79, 73-110	2023
Searching for Rigidity in Algebraic Starscapes J. Math. Arts. 16, 57-74 With C. Xu. Cover Article.	2022

Algebraic and Arithmetic Geometry

Untilting Line Bundles on Perfectoid Spaces Int. Math. Res. Not. IMRN (3):2572-2591	2023
On Picard Groups of Perfectoid Covers of Toric Varieties Eur. J. Math. 9, 16 With A. Ray and P. Wear	2023
Projective Geometry for Perfectoid Spaces Münster J. Math. 14, 445-484	2021

Artistic Exhibitions and Publications

Cubic Surface Exhibited at Maison Poincaré for <i>Creation: Between Art and Mathematics</i> With C. Gómez-González and S. Trettel	2026
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Twenty-Seven	2025
w/k - Between Science and Art: Mathematical Sculptures, ed. M. Damrau and M. Skrodzki With <i>D. Rostamloo</i>	
Bohemian Eigenvalue Starscape No. 4 and No. 7	2024
Catalog of the 2024 Joint Math Meetings Art Exhibition With <i>E. Brown</i>	
Rigid Algebraic Starscapes	2022
Exhibited at the MAA Golden Section Art Exhibition With <i>C. Xu</i>	
The Fabric of Spacetime	2020
Illustrating Mathematics, ed. D. Davis, American Mathematical Soc., 16-17 Exhibited at DXARTS for <i>Machines of Survival</i> With M. Maynard. Cover Article.	
3D Printing and Fractal Deformations	2020
Illustrating Mathematics, ed. D. Davis, American Mathematical Soc., 100-101	

GRANTS AND FUNDED PROJECTS

Mathematical Illustration Apprentice Program	2025-2026
External: Matheneum Foundation \$50,000 Principal Investigator Funding and programming for visits to the Institut Henri Poincaré for early career scientists	
Research enhancement grant for primarily undergraduate institute faculty	2024-2027
External: American Mathematical Society - Simons Foundation \$10,000 Principal Investigator	
Tangible Calculus: Using Models and Manipulatives to Enhance Conceptual Understanding	2025
Internal: St. Lawrence University Curricular Innovation Grant and Instructional Mini-Grant \$4,000 Co-Principal Investigator with D. Condon, P. Lock, D. Look Funding to design and fabricate interactive models and activities for Calculus III in the makerspace	

UNDERGRADUATE MENTORSHIP

I regularly mentor undergraduate research in pure mathematics, cryptography, visualization, mathematical art, fabrication, and computational mathematics through honors theses, summer fellowships, independent studies, and collaborative research projects. Many projects have led to conference presentations, exhibitions, publications, or ongoing collaborations. Since co-founding Explore Lab Makerspace, I have supervised many projects which have made extensive use of the space.

St. Lawrence University Mentoring

Cordelia Sherwood: Securing Motorsport Telemetry with AES Cryptography	2025-2026
<i>Honors SYE</i> Explored ring theory, cryptography and electronics to program AES into an RC car.	
Charis Bouton: Building Intuition Through Models of Solids of Revolution	2025
<i>Honors SYE</i> Used the makerspace to design models and activities for our course in integral calculus.	

Carter Banks: Visualizing Group Actions on Platonic Solids <i>McNair Research Fellowship and SYE</i> Used the makerspace to build models depicting group actions with moving lights.	2025
Allison House: Linear Algebra through Geology <i>SLU Fellowship</i> Designed activities exploring geological applications for our Linear Algebra course.	2025
Elise Heppel: Divisibility Properties of Reversed Digit Pairs <i>SLU Fellowship</i> Conducted original number theory research presented at the MAA Seaway and Joint Math Meetings.	2025
Ally McCormick: Visualizing Geometric Concepts through Light and Reflection <i>Honors SYE</i> Used the makerspace to build an infinity box of the octahedron to see its noneuclidean tiling space.	2025
Tye Royal: Building an Enigma Machine <i>SLU Fellowship and Honors SYE</i> Used the makerspace and programmed microcontrollers to build a functioning enigma machine.	2023-2025
Kobe Villeneuve: Graph Centrality's Impact on Economic Welfare <i>Honors SYE</i> Conducted an interdisciplinary investigation of graph theory and economics, in preparation for publication.	2025
Zijian Zhang: Galois Theory and the Insolvability of the Quintic <i>Honors SYE</i> Wrote an expository piece on the classical use of Galois theory to prove the insolvability of the quintic.	2025
Charlie Gartner: A Tripartite Elliptic Curve Diffie-Hellman <i>SYE</i> Programmed and tested an elliptic curve cryptosystem in sagemath.	2025
Molly Sullivan: A Comic on Moduli Spaces <i>Summer research fellowship</i> Used digital art and humor to explain moduli theory to a general audience.	2024
Eliza Brown: 3D Printing Escape Velocities in the Mandelbrot Set <i>SLU Fellowship and Honors SYE</i> Used the makerspace to 3D print visualizations of fractals. Presented at the art show of the ICERM Illustrating Math Reunion/Expansion in August 2025	2024
Ryann Murray: Can You Hear the Shape of an Instrument? <i>Honors SYE</i> Used the makerspace to build a <i>fractal</i> ukelele.	2024
Robert Nordberg: Fractal Antennae <i>SYE</i> Used the makerspace and costume shop to make clothes with wearable electronic elements.	2024
Sarah Bellefleur: Shor's Algorithm and Quantum Cryptography <i>Honors SYE</i> Wrote an expository piece describing how to use quantum computers to factor large numbers.	2023-2024
Eliza Brown: Starscapes of Eigenvalues of Bohemian Matrices <i>SLU Fellowship</i> Used high performance computing to create art from eigenvalues of integer matrices. Presented and published at the 2024 Joint Math Meetings.	2023

Yeyuxi Yi: Tribinomials, Fibunomials, and Generalizations

2023

Saints Start Fellowship

Computed experimental results on certain combinatorial sequences.

Berkeley Mentorship

Daniel Rostamloo: Twenty-Seven

2021-2025

Created of an interactive sculpture of a cubic surface and its 27 lines.

Presented at the meeting of the German Mathematical Society and published in w/k: Between Science and Art

Florence Lam: Implementing Boneh-Franklin Identity Based Encryption

2021-2023

Senior Thesis

Published in SIURO.

Candyce Xu: Algebraic Starscapes and Galois Theory

2020-2022

Senior Thesis

A visual and experimental project at the intersection of number theory and digital art.

Shown in the MAA Golden Section and Bridges art exhibitions. Published in J. Math. Arts.

TEACHING

My teaching emphasizes inquiry driven mathematics, mathematical communication, undergraduate research integration, mathematical visualization, and computational experimentation. I regularly teach proof based courses in algebra and cryptography alongside courses in linear algebra and calculus, and integrate written and oral communication projects like chalktalks.

St. Lawrence University Teaching

Fall 2025

Math 316: Ring Theory

Math 217: Linear Algebra (2 Sections)

Spring 2024

Math 391/CS 391: Cryptography

Math 217: Linear Algebra (2 Sections)

Fall 2024

Math 217: Linear Algebra (2 Sections)

Spring 2023

Math 315: Group Theory

Math 217: Linear Algebra

Fall 2023

Math 205: Multivariable Calculus

Math 217: Linear Algebra (2 Sections)

Spring 2023

Math 391/CS 391: Cryptography

Math 217: Linear Algebra (2 Sections)

Fall 2022

Math 205: Multivariable Calculus (2 Sections)

Math 217: Linear Algebra

Pre St. Lawrence Teaching

UC Berkeley

2020-2022

Lead Instructor: Abstract Algebra, Cryptography

University of Washington

2013-2019

Lead Instructor: Precalculus, Calculus I-IV, Cryptography

Teaching Assistant: Business Calculus, Calculus I-III, Abstract Algebra for Teachers

ACADEMIC LEADERSHIP AND INITIATIVES

I work to support interdisciplinary approaches to scholarship, pedagogy, and outreach at institutional and international scales.

Intitutional Initiatives

Co-Founder and Co-Director: The Explore Lab Makerspace

2023-

Established and manage a makerspace supporting student and faculty scholarship and exploration.

Secured internal and external funding for equipment, maintenance, and student staff.

Coordinate class visits to use the space for project components.

With K. Angstadt

Disciplinary Initiatives

Trimester Research Program Organizer: Illustration as a Mathematical Research Technique

2026

Institut Henri Poincaré, Paris.

Conceptualized, proposed, and fully coordinated a trimester long research program.

Managed scheduling, fundraising, and invitations for 40 long term visitors and 250 short term visitors.

SERVICE

Institutional and Departmental Service

Faculty Development Committee

2023-

St. Lawrence College Democrats, Faculty Advisor

2023-

Statistics/Applied Mathematics Search Committee Co-Chair

2025

Admitted Students Day Admissions Panel

2025

Served a panel for parents of admitted students, presenting on the Explore Lab.

Pi Mu Epsilon Faculty Advisor

2022-2025

Served as advisor for the St. Lawrence Chapter of the national mathematics honor society.

In Development Podcast

2024

Hosted and produced an episode on reducing bias in grading.

Presenter, May College

2023

Served on a faculty panel on innovative teaching

Presented on *Revise and Resubmit* for sophomore level math courses.

Disciplinary Service

Illustrating Mathematics Seminar Online

2023-

Co-founded and organize an internationally attended monthly seminar averaging 50-100 attendees

Rigorous Illustrations	2026
Organizer for week long conference at the Institut Henri Poincaré	
MAA Seaway Section Workshop	2025
Organized and led participatory workshop about fabricating models for calculus courses.	
MRC Special Session at the Joint Mathematics Meetings	2024
Session organizer	
Construct3D: 3D Printing and Digital Fabrication for Students and Educators	2023
Higher education programming committee	
MAA Golden Section Art Show	2022
Organizer and curator	
Berkeley RTG Seminar in Arithmetic Geometry	2020-2022
Organizer of weekly invited speaker seminar	
Illustrating Mathematics Graduate Summer School	2021
Organizer of week long summer school at the Parks City Mathematics Institute	
Berkeley Undergraduate Number Theory Conference	2020
Organizer of weekend online conference	
ICERM Graduate Student and Postdoc Seminar	2019
Organizer of weekly internal seminar	
Peer Reviewer	2019-
Referee Bridges, Proc. Symp. Appl. Math, J. Math. Arts, Simons Symposium Proceedings, Abh. Math. Semin. Univ. Hambg, HJM, Open Mathematics, and others.	

INVITED TALKS.

Keynote talks

MAA Seaway Section	2024
Mathematics Conference and Competition of Northern New York	2024

Invited seminar talks

Algebraic Geometry @ PUI Seminar	2024
University of Utah Algebraic Geometry Seminar	2024
Bay Area Mathematical Adventurers	2022
A joint lecture series from San Jose State and Santa Clara University	
Santa Clara University Colloquium	2022
St. Lawrence University Colloquium	2021
University of Utah Number Theory Seminar	2021
UC Berkeley RTG Seminar in Arithmetic Geometry	2020
Brown University Algebraic Geometry Seminar	2019
Dartmouth College Algebra and Number Theory Seminar	2019

University of Washington Algebra and Algebraic Geometry Seminar	2019
Boston University Number Theory Seminar	2018
Columbia University Number Theory Seminar	2018
Rice University Algebra and Number Theory Seminar	2018
University of Arizona Algebraic Geometry Seminar	2018
Invited conference talks	
Integrating Research and Illustration in Number Theory Institut Henri Poincaré, Paris	2026
ICERM Illustrating Math Reunion/Expansion	2025
Joint Math Meetings AMS Special Session on Mathematics and Computation	2025
Annual Meeting of the German Mathematical Society Symposium on Mathematics and the Arts	2023
Construct 3D Workshop on 3D Printing Static Animations	2023
25th International Linear Algebra Society: Madrid Symposium on Bohemian Matrix Theory	2023
Casa Matemática Oaxaca / Banff International Research Station Modern Advances in Mixed Characteristic Commutative Algebra	2022
Joint Math Meetings Mini Course on 3D Printing in Mathematics	2022
Berkeley Undergraduate Number Theory Conference	2020